BREAKING THE DEFENDERS

Measuring Defense Impairment at Scale Through MSP Telemetry

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Defense Evasion
~~TA0005~~ (retired)

Stealth
TA0005

Defense Impairment
TA0112



The attacker goes beyond hiding and directly interferes with security controls, reducing their ability to prevent, detect or respond.

Why?

The impairment attempt may be the last observable signal before defenders go blind.



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What?

Behaviors targeting defensive controls revealed through MSP telemetry across ransomware, espionage, and other intrusions.



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How?

Map adversary behavior, identify visibility gaps, and measure real detection coverage.



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Telemetry
Intelligence

What defensive capability does the adversary intend to affect?

BLIND	WEAKEN	MISLEAD	BLOCK
Remove visibility	Bypass or disable protection	Corrupt evidence or abuse trusted controls	Interfere with response and recovery

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CAMBODIA-FOCUSED CLUSTER, 2026

One intrusion used five ways to reduce visibility

01	02	03	04	05
ENUMERATE	**EXCLUDE**	**PATCH**	**KILL**	**DEGRADE**
Find the controls	Create blind spots	Reduce telemetry	Stop processes	Remove capability

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CAMBODIA-FOCUSED CLUSTER

Signed does not mean safe

ardrv.sys

SIGNED DRIVER

KERNEL ACCESS

PROCESS TERMINATION

Kernel termination with a user-mode fallback

DEFENSE IMPAIRMENT BEYOND PROCESS TERMINATION

A control can keep running and still be impaired

RUNNING does not guarantee **EFFECTIVE**

Privileges removed

Integrity lowered

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NOT EVERY VISIBILITY GAP BEGINS WITH A DISABLED CONTROL

Khmer Shadow bypassed userland monitoring

01

RESTORE NTDLL

Replace the hooked .text section with a clean copy from disk

02

DIRECT SYSCALLS

Hell's Gate resolves syscall numbers and calls the kernel directly

The agent can remain active while userland interception points disappear



RED HERON / SIXZUT

The host can lie (unlike the hips)

LD_PRELOAD

FILES HIDDEN

PROCESSES HIDDEN

CONNECTIONS HIDDEN

No visible evidence

does not prove

NO ACTIVITY

RANSOMWARE / PATTERN ACROSS CASES

Ransomware changes the defensive stack before encryption

01	**CHANGE SETTINGS**	Policies, exclusions and firewall
02	**STOP SERVICES**	Security, backup and virtualization
03	**KILL PROCESSES**	User-mode tools or vulnerable drivers
04	**ERASE SIGNALS**	ETW, event logs and command history


APT
RANSOMWARE

The last signal before darkness

INTERACTION

FAILURE

GAP

OUTCOME

The attempt can remain valuable even when it succeeds

DET0900 / AN2038

Three values turn DET0900 into a working detection

MUTABLE ELEMENT	QUESTION THE DETECTION MUST ANSWER
ProtectedServiceList	Which security controls must remain healthy?
CrashCorrelationWindow	How close must interaction and failure be?
TelemetryGapThreshold	How much silence is abnormal?

For MSPs: Tune by **tenant**, **product** and **reporting cadence**

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When one sensor disappears, look sideways



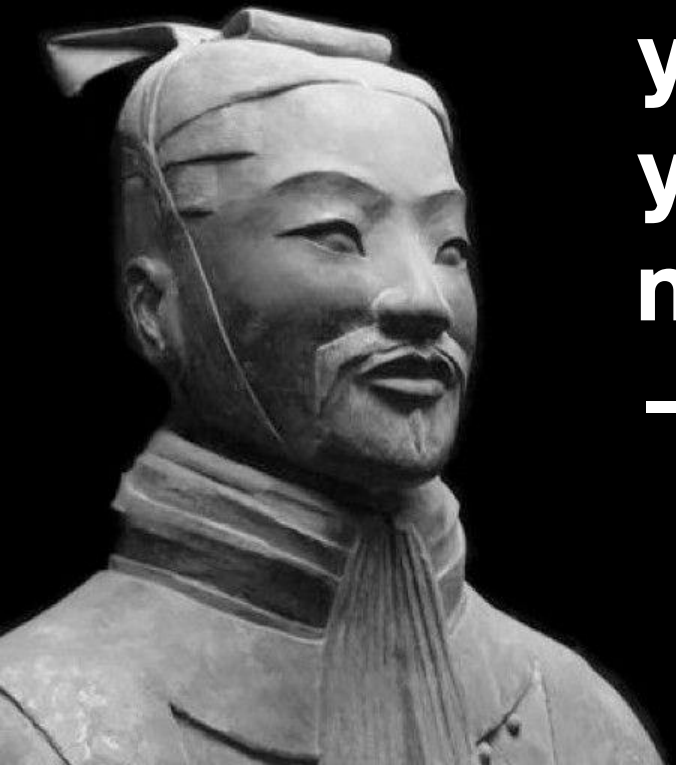
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Coverage is not a percentage

Can we see the attempt?

Can we see the consequence?

Can we detect the gap?



“Know your enemy, know yourself, and make sure your enemy’s EDR knows nothing.”

— Sun Tzu, Windows Event ID 7036

Resources and contact



TECHNICAL REFERENCES, NOTES, AND SUPPORTING MATERIAL

github.com/spontiroli/
breaking-the-defenders

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